TOWARD THE TWIN PRIME CONJECTURE: CERTIFICATES, LIMITS OF THE SIEVE, AND A WORKING PROGRAMME

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ABSTRACT. This is a living working paper. It does *not* prove the twin prime conjecture, and it does not improve the best known bound on small gaps between primes. It records (i) the state of the problem as of the date above, (ii) what this project has verified for itself, by exact computation and by the Lean 4 kernel, (iii) small results of our own: a no-go statement for a natural attempt to sharpen the pigeonhole step, on the plain and on the ε-enlarged simplex, and a generating-function formula that makes certified computation with all symmetric test functions cheap, and (iv) a programme of original directions that later versions will test. Every statement carries one of the labels LEAN-CHECKED, COMPUTED, PROVED HERE, LITERATURE, CLAIMED (a preprint not yet independently verified), or OPEN.

1. THE PROBLEM AND THE HONEST POSITION

Write p_n for the n-th prime and $H_1 = \liminf_{n\to\infty}(p_{n+1} - p_n)$. The twin prime conjecture is the statement $H_1 = 2$. It is OPEN.

A finite set $\mathcal{H} = \{h_1 < \cdots < h_k\}$ of integers is *admissible* if for every prime p it misses at least one residue class modulo p. Let DHL$[k, 2]$ be the statement that every admissible k-tuple has infinitely many translates $n + \mathcal{H}$ containing at least two primes, and let $H(k)$ be the least diameter $h_k - h_1$ of an admissible k-tuple. Then

$$\text{DHL}[k, 2] \implies H_1 \le H(k), \tag{1}$$

and DHL$[2, 2]$ applied to $\{0, 2\}$ is the twin prime conjecture. All progress on H_1 since 2013 consists of proving DHL$[k, 2]$ for smaller k.

Where the bound stands (12 November 2026).

$H_1 \le$	k	Source	Label
7×10^7	3.5×10^6	Zhang [18]	LITERATURE
600	105	Maynard [10]	LITERATURE
246	50	Polymath8b [13]	LITERATURE
240	49*	Stadlmann [16]	CLAIMED
236	48*	Kintali [8]; Song–Yue [15]	CLAIMED
212	45	Charton et al. [3]	CLAIMED
186	40	OpenAI [12]	CLAIMED
12	5	Maynard, assuming EH [10]	LITERATURE
6	3	Polymath8b, assuming GEH [13]	LITERATURE

* Value of k inferred by us from $H(49) = 240$ and $H(48) = 236$, not read from the source.

The bound 246 stood from 2014 until the last days of August 2026. It has been formalised in Lean 4, assuming only the Bombieri–Vinogradov theorem [1]. The four bounds below it in

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the table appeared within one week. They rest on Stadlmann's idea of enlarging the support of the sieve weights using equidistribution estimates for moduli with a large smooth factor. The Lean development accompanying the bound 186 is conditional on three stated inputs (two exponential-sum bounds and a family of numerical integral bounds) [12]. As of 7 September 2026 a standard reference still listed 246 as the best fully established value [11]; on the date of this version we found no report of complete independent verification of the 2026 claims and no claimed bound below 186. We therefore keep two columns throughout: *established* (246) and *claimed* (186).

Why 2 is a different problem. Under the generalised Elliott–Halberstam conjecture the sieve gives $H_1 \le 6$, and Polymath8b showed that 6 cannot be improved by purely sieve-theoretic arguments: this is Selberg's parity obstruction [13]. Lowering the bound (Track A below) and proving the conjecture (Track B) are therefore different tasks. No amount of success on Track A reaches 2.

2. What this project has verified for itself

2.1. **The combinatorial half, checked by the Lean kernel.** Our Lean development is written in core Lean 4, without Mathlib. It defines primality, the twin prime conjecture, the predicate `BoundedGaps B` (infinitely many pairs of distinct primes at distance at most B), admissibility, and DHL$[k, 2]$. It contains no `sorry`, and the only axioms it uses are `propext` and `Quot.sound`. The statement DHL$[k, 2]$ is always a hypothesis, never an axiom.

Theorem 1 (Lean-checked). (1) *The twin prime conjecture is equivalent to* ***`BoundedGaps 2`****.*

(2) *If a finite Boolean check succeeds on a list $\mathcal{H}$ (it misses a class modulo every m with $2 \le m \le |\mathcal{H}|$), then $\mathcal{H}$ is admissible. The moduli $p > |\mathcal{H}|$ are handled by a formal pigeonhole argument.*

(3) *For each pair (k, D) in Table 1, for $(k, D) = (56, 278)$, $(57, 282)$, $(58, 288)$, $(59, 300)$, $(60, 304)$, $(61, 310)$, $(62, 320)$, $(63, 324)$, $(64, 330)$, and for $(102, 578)$ and $(105, 600)$, there is an explicit admissible k-tuple in $[0, D]$, and* DHL$[k, 2]$ *implies* ***`BoundedGaps D`****. In particular* DHL$[2, 2]$ *implies the twin prime conjecture.*

Table 1. Pairs (k, D) with a kernel-checked admissible k-tuple of diameter D. The tuples were found by our own search; each D agrees with the value of $H(k)$ in published tables, whose optimality we have not re-proved.

k	D	k	D	k	D	k	D	k	D	k	D
2	2	11	36	20	80	29	130	38	176	47	226
3	6	12	42	21	84	30	136	39	182	48	236
4	8	13	48	22	90	31	140	40	186	49	240
5	12	14	50	23	94	32	146	41	188	50	246
6	16	15	56	24	100	33	152	42	196	51	252
7	20	16	60	25	110	34	156	43	200	52	254
8	26	17	66	26	114	35	158	44	210	53	264
9	30	18	70	27	120	36	162	45	212	54	270
10	32	19	76	28	126	37	168	46	216	55	272

Theorem 1 is elementary. Its value is that implication (1) is now a checked interface: any future proof of DHL$[k,2]$, by anyone, yields the corresponding gap bound with nothing left to trust on the combinatorial side.

2.2. **The sieve constant, with exact certificates.** Let $\mathcal{R}_k = \{t \in [0,1]^k : \ t_1 + \cdots + t_k \le 1\}$ and, for square-integrable F supported on $\mathcal{R}_k$,

$$I_k(F) = \int_{\mathcal{R}_k} F^2, \qquad J_k^{(m)}(F) = \int \left(\int F \, dt_m \right)^2 \prod_{i \ne m} dt_i, \qquad M_k = \sup_F \frac{\sum_{m=1}^k J_k^{(m)}(F)}{I_k(F)}.$$

Maynard's theorem [10, Prop. 4.2] states that if the primes have level of distribution θ and $\theta M_k/2 > 1$ then DHL$[k,2]$ holds. With Bombieri–Vinogradov ($\theta < 1/2$) the condition is $M_k > 4$.

We take $F = \sum_{a+2b \le d} c_{a,b} (1-P_1)^a P_2^b$ with $P_j = \sum_i t_i^j$, as in [10, §8]. All integrals reduce to

$$\int_{\mathcal{R}_k} (1-P_1)^A P_2^B \, dt = \frac{A!}{(k+A+2B)!} \, G_k(B), \qquad \sum_{B \ge 0} G_k(B) \frac{x^B}{B!} = \left(\sum_{b \ge 0} \frac{(2b)!}{b!} x^b \right)^k,$$

which follows from the Dirichlet integral. The two quadratic forms are then exact rational matrices. We compute an approximate top eigenvector in floating point, round it to rationals, and evaluate the Rayleigh quotient in exact arithmetic; the last step is a rigorous lower bound whatever happened before it.

Theorem 2 (Computed, exact rational arithmetic)**.** *With F a polynomial in $1-P_1$ and P_2 only,*

$$M_{105} \ge 4.00206976, \qquad M_{102} \ge 4.00271343, \qquad M_5 \ge 2.00410541.$$

The first value reproduces Maynard's $M_{105} > 4$ with $d = 11$ (42 coefficients). The third gives DHL$[5,2]$ and $H_1 \le 12$ under the Elliott–Halberstam conjecture, again reproducing [10]. For $d = 11, 15, 19$ the least k at which this family certifies $M_k > 4$ is $105, 102, 102$, and at $k = 101$ it gives only 3.99931 ($d = 19$): polynomials in P_1, P_2 alone stop at $k = 102$.

2.3. **All symmetric test functions.** To go further we allow every symmetric polynomial. Since $P_1, \ldots, P_k$ generate the symmetric polynomials, the functions

$$(1-P_1)^a \, P^b := (1-P_1)^a \prod_{j \ge 2} P_j^{b_j}, \qquad a + w(b) \le d, \quad w(b) = \sum_j j b_j,$$

span all symmetric polynomials of degree at most d (for $k \ge d$). The integrals have a closed form that we have not seen stated in this way (Proved here; possibly folklore).

Lemma 3. *Let $X_1, \ldots, X_k$ be independent exponential random variables of mean 1 and $E_k(b) = \mathbb{E} \prod_j P_j(X)^{b_j}$. Then*

$$\int_{\mathcal{R}_k} (1-P_1)^A P^b \, dt = \frac{A!}{(k+A+w(b))!} \, E_k(b), \qquad \sum_b E_k(b) \frac{s^b}{b!} = \left(\sum_c w(c)! \, \frac{s^c}{c!} \right)^k,$$

where $s^b = \prod_j s_j^{b_j}$ and $b! = \prod_j b_j!$. In particular each $E_k(b)$ is a positive integer, computed exactly by multiplying exponential generating functions.

Proof. For a monomial, the Dirichlet integral gives $\int_{\mathcal{R}_k} (1-P_1)^A t^\alpha \, dt = A! \, \alpha! / (k+A+|\alpha|)!$, and $\alpha! = \mathbb{E} X^\alpha$. Summing over the monomials of P^b, all of degree $w(b)$, gives the first identity. For the second, $\mathbb{E} \exp(\sum_j s_j P_j(X)) = \left(\mathbb{E} \exp(\sum_j s_j X_1^j) \right)^k$ by independence, and $\mathbb{E} \exp(\sum_j s_j X_1^j) = \sum_c \mathbb{E} X_1^{w(c)} \, s^c/c! = \sum_c w(c)! \, s^c/c!$, as formal power series. □

The J-functional reduces to the same lemma in $k-1$ variables: with $P_j = P'_j + t_k^j$,

$$\int_0^{1-P'_1} (1-P_1)^a P^b \, dt_k = \sum_{e \le b} \binom{b}{e} \frac{a!\, w(e)!}{(a+w(e)+1)!} (1-P'_1)^{a+w(e)+1} P'^{\,b-e}.$$

So both Gram matrices are exact rational matrices. We find an approximate top eigenvector by inverse iteration with a fixed shift above $\frac{k}{k-1}\log k$, which can only converge to the top eigenvalue. We take its floating-point midpoints as exact dyadic rationals and evaluate the Rayleigh quotient exactly.

Theorem 4 (Computed, exact rational arithmetic)**.**

$$M_{105} \ge 4.17427, \quad M_{64} \ge 4.00363, \quad M_{58} \ge 4.00927, \quad M_{56} \ge 4.00519.$$

The first two use all symmetric polynomials of degree 11 *and* 12 *(*195 *and* 272 *coefficients). The last two use the subspace spanned by* $(1-P_1)^a P^b$ *with* b *supported on even* j*, as in* [13, §7]*, of degree* 16 *and* 18 *(*307 *and* 471 *coefficients).*

Basis	degree	coefficients	least k with certified $M_k > 4$
$1-P_1$, P_2	19	110	102
all symmetric	12	272	64
even power sums	16	307	58
even power sums	18	471	≤ 56

Corollary 5 (end-to-end within this project)**.** *Assuming the Bombieri–Vinogradov theorem and* [10, Prop. 4.2]*, both* Literature*, Theorems 4 and 1 give* $H_1 \le 278$*.*

Polymath8b reached $M_{54} > 4.00238$ with even-part polynomials of degree 23 [13, Thm. 3.9], and hence $H_1 \le 270$. Our pipeline is two steps of k short of that, and the tuple for $k = 56$ came from our own search and may not be the narrowest. Corollary 5 remains far weaker than the literature. It is the position of our own pipeline on the ladder $600 \to 270 \to 246 \to 240 \to 212 \to 186$, which we intend to climb rung by rung.

2.4. **What the plain method cannot do.** Polymath8b proved $M_k \le \frac{k}{k-1}\log k$ [13]. Hence (Computed from that bound) $M_k > 4$ forces $k \ge 51$, and on the plain simplex DHL$[k,2]$ requires a level of distribution $\theta > 2(k-1)/(k\log k)$:

k	50	45	40	35	30	20	10	5
$\theta >$	0.501	0.514	0.529	0.547	0.568	0.634	0.782	0.994

The 2026 results go below $k = 50$ by changing the support of F, not by proving a uniform level of distribution of this size.

3. Results of our own: rigidity of the pigeonhole step

Every bound above ends with the same step: if the weighted average number of primes among $n+h_1, \dots, n+h_k$ exceeds 1, two of them are prime, and they are at most $h_k - h_1$ apart. A natural hope is to do better with unequal weights. Fix D, and give each position a weight $c_i \ge 0$. Call a set of positions D-*separated* if any two of its shifts differ by more than D, and let $\alpha_D(c)$ be the largest total weight of a D-separated set. If no two primes among the $n+h_i$ are within D, then $\sum_i c_i \mathbf{1}_{n+h_i \text{ prime}} \le \alpha_D(c)$. Writing

$$M_k[c] = \sup_F \frac{\sum_m c_m J_k^{(m)}(F)}{I_k(F)},$$

the sieve therefore yields $H_1 \le D$ as soon as $\frac{\theta}{2} M_k[c] > \alpha_D(c)$. For $c = (1, \dots, 1)$ and $D = h_k - h_1$ this is the usual criterion. Could a cleverer c certify some $D < h_k - h_1$ that no sub-tuple certifies?

Proposition 6 (Proved here; possibly folklore)**.** *Let ω be the largest number of shifts h_i contained in an interval of length D. Then for every $c \ge 0$, not identically zero,*

$$\frac{M_k[c]}{\alpha_D(c)} \le M_\omega.$$

So the weighted criterion never certifies more than the standard criterion applied to the best sub-tuple of diameter at most D.

Proof. Let G be the graph on $\{1, \dots, k\}$ joining $i \ne j$ when $|h_i - h_j| \le D$. Its stable sets are the D-separated sets and its cliques are the sets of shifts lying in an interval of length D. Normalise $\alpha_D(c) = 1$, so c lies in

$$P = \{c \ge 0 : \ c(S) \le 1 \text{ for every stable set } S\}.$$

The graph G is a unit interval graph, hence perfect, and so is its complement by the perfect graph theorem [9]. For a perfect graph the polytope cut out by the clique inequalities is the convex hull of the indicator vectors of its stable sets [5]. Applying this to the complement of G gives

$$P = \operatorname{conv}\{\mathbf{1}_K : \ K \text{ a clique of } G\}.$$

The map $c \mapsto M_k[c]$ is a supremum of linear functionals, hence convex, and therefore $M_k[c] \le \max_K M_k[\mathbf{1}_K]$. It remains to show $M_k[\mathbf{1}_K] \le M_j$ for $|K| = j$; this suffices because M_j is non-decreasing in j (adjoin a variable and concentrate F near its zero set). Write $t = (s, u)$ with s the coordinates in K and u the others, put $\sigma(u) = 1 - \sum u \ge 0$, and let $F_u = F(\cdot, u)$, a function on the scaled simplex $\sigma(u)\mathcal{R}_j$. For $m \in K$ we have $J_k^{(m)}(F) = \int J_j^{(m)}(F_u)\, du$, and $I_k(F) = \int I_j(F_u)\, du$. Rescaling F_u to $\mathcal{R}_j$ multiplies I_j by σ^j and each $J_j^{(m)}$ by σ^{j+1}, so

$$\sum_{m \in K} J_j^{(m)}(F_u) \ \le \ \sigma(u)\, M_j\, I_j(F_u) \ \le \ M_j\, I_j(F_u).$$

Integrating over u gives the claim. □

Remark 7 (Computed, consistency check for $k = 3$). With unrestricted polynomials of degree 6 on $\mathcal{R}_3$ we find the lower bounds $M_3[(1,1,1)] \ge 1.6464$, $M_3[(1,1,0)] \ge 1.3579$ and $M_3[(\frac{1}{2}, 1, \frac{1}{2})] \ge 1.1604$, against $M_2 = 1/(1 - W(1/e)) = 1.3859\ldots$. The half-weighted functional is far below M_2, as the proposition requires.

The same rigidity on the enlarged simplex. The ε-enlarged simplex of Polymath8b [13, Thm. 3.12] takes F supported on $(1+\varepsilon)\mathcal{R}_k$ and restricts the outer integral of each $J^{(m)}$ to $(1-\varepsilon)\mathcal{R}_{k-1}$. Call the resulting constant $M_{k,\varepsilon}$. The criterion $M_{k,\varepsilon} > 2/\theta$ is valid in two cases (Literature): under EH[θ] when $1 + \varepsilon < 1/\theta$, and under GEH[$\theta$] when $\varepsilon < 1/(k-1)$. With Bombieri–Vinogradov the first case allows every $\varepsilon < 1$, and Polymath8b's $H_1 \le 246$ uses $M_{50,1/25} > 4.0043$ [13, Thm. 3.13]. Unequal weights are no more useful there.

Proposition 8 (Proved here; possibly folklore)**.** *For $0 \le \varepsilon < 1$ and every weight $c \ge 0$, $c \ne 0$, $M_{k,\varepsilon}[c]/\alpha_D(c) \le M_{\omega,\varepsilon}$, with equality for the indicator of a largest clique. Since the conditions on ε above do not get harder as k decreases, the weighted criterion never certifies more than the standard one for the best sub-tuple of diameter at most D.*

Proof. As for Proposition 6, it suffices to show $\sum_{m \in K} J_{k,\varepsilon}^{(m)}(F) \le M_{j,\varepsilon} I_k(F)$ for a clique K of size j, together with $M_{j,\varepsilon} \le M_{\omega,\varepsilon}$. Fix the coordinates u outside K, with $|u| \le 1 - \varepsilon$ (other slices do not meet the J-region). The slice F_u is supported on $(1 + \varepsilon - |u|)\mathcal{R}_j$, and its J-region

is $(1-\varepsilon-|u|)\mathcal{R}_{j-1}$. Put $\mu = (1+\varepsilon-|u|)/(1+\varepsilon) \le 1$ and $H(x) = F_u(\mu x)$, which is supported on $(1+\varepsilon)\mathcal{R}_j$. Then $I_j(F_u) = \mu^j I_j(H)$, and each slice J-term equals μ^{j+1} times the integral of $(\int H\, dx_m)^2$ over $|x'| \le (1-\varepsilon-|u|)/\mu$. Since

$$\frac{(1-\varepsilon-|u|)(1+\varepsilon)}{1+\varepsilon-|u|} \le 1-\varepsilon \iff -|u|(1+\varepsilon) \le -|u|(1-\varepsilon),$$

that region lies inside $(1-\varepsilon)\mathcal{R}_{j-1}$, so the slice contributes at most $\mu^{j+1} M_{j,\varepsilon} I_j(H) = \mu M_{j,\varepsilon} I_j(F_u)$. For $j = 1$ use Cauchy–Schwarz, which gives $M_{1,\varepsilon} = 1+\varepsilon$. Integrate over u and use $\mu \le 1$. Monotonicity in j follows by embedding: take $F(s,u) = H(s/\mu_\delta)\mathbf{1}_{[0,\delta]^{k-j}}(u)$ with $\mu_\delta = 1-(k-j)\delta/(1+\varepsilon)$ and let $\delta \to 0$. □

Slicing naively gives each slice its own parameter $\varepsilon' = \varepsilon/(1-|u|) \ge \varepsilon$, which would only bound the weighted constant by $\sup_{\varepsilon' \ge \varepsilon} M_{j,\varepsilon'}$, a weaker statement. The single rescaling by μ above avoids that loss.

Remark 9 (Computed, $k = 3$). With unrestricted polynomials (degree 5 for $k = 3$, degree 10 for $k = 2$, cut off at $|t| \le \rho$ for $\rho \in \{1-\varepsilon, 1, 1+\varepsilon\}$):

ε	$M_{2,\varepsilon} \ge$	$M_{3,\varepsilon}[(1,1,0)] \ge$	$M_{3,\varepsilon}[(\frac{1}{2},1,\frac{1}{2})] \ge$	$M_{3,\varepsilon} \ge$
0	1.385933	1.349844	1.160121	1.646427
1/10	1.518567	1.482143	1.264698	1.793492
1/5	1.627161	1.585674	1.341926	1.882282
1/3	1.717104	1.663743	1.413309	1.917001

Both weighted columns stay below $M_{2,\varepsilon}$, as the proposition requires.

Remark 10 (scope). Neither proposition covers more general polytope supports [13, §3], Stadlmann's hybrid supports [16], J-regions that depend on m, or arguments that are not linear in the indicators of primality. Those are Open.

4. Programme

Track A: lowering the bound.

(1) *Climb the ladder.* Push the certified pipeline from $k = 56$ to $M_{54} > 4$ ($H_1 \le 270$; needs degree about 20 to 23), then add the ε-enlarged simplex ($k = 50, 246$), then Stadlmann's region ($k = 49, 240$).
(2) *Floor of the method (our main original direction).* For the plain simplex the dual bound $M_k \le \frac{k}{k-1}\log k$ shows exactly where the method stops. No analogous upper bound is known to us for the 2026 hybrid supports. A rigorous dual certificate there would say whether 186 is close to the limit of the present inputs or far from it, and which input (which Type I/II/III range) is binding.
(3) *Which rung is cheap.* From Table 1, the gain per unit of k below the claimed frontier is $186 \to 182 \to 176 \to 168 \to 162 \to 158$ for $k = 40, \dots, 35$: the steps $39 \to 38$ and $38 \to 37$ are worth the most.

Problem 11 (Open)**.** Give a rigorous upper bound for the variational quantity of [16] as a function of k and of the admissible exponents, and deduce the least k the hybrid method can reach with the currently proved estimates.

Track B: the parity barrier. No sieve of the above kind can distinguish integers with an even number of prime factors from those with an odd number. Known ways around the barrier all inject bilinear or structural information. We will study, one per session, what each would need in order to say something about twin primes: Chen's theorem [4]; Bombieri's asymptotic sieve [2]; the parity-breaking sieve of Friedlander and Iwaniec [6]; Heath-Brown's theorem that Siegel zeros would imply the conjecture [7]; the function-field proof of Sawin and Shusterman [14]; and the entropy-based results on correlations of multiplicative functions [17].

Problem 12 (Open, to be made precise in Lean)**.** State, as a single explicit hypothesis on bilinear sums over $n(n+2)$, the weakest known input that implies the twin prime conjecture, and formalise the implication.

Rules of this paper. A numerical bound enters the "established by this project" column only with an exact certificate and, for the combinatorial part, a Lean proof. Ideas are logged with their outcome, including failures. Nothing here is peer reviewed.

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